

Introductions often serve as as confessions, the more vulnerable of which neither beg for forgiveness nor serve apologetics, remaining virtuous inasmuch as one's good faith efforts under honest self reflective expression are for *you*, the reader. The empirical tradition, formal or otherwise, may be similarly and somewhat reductively read as an ongoing attempt to engage in such narrative leaps of faith; we task ourselves to convey how the world appears generally, often towards a law like description, whilst grappling with the inescapable fact that our perspective must start "from somewhere".

The explication of this tension has a long history, typically played out under an assumed irreducible dialectic between universalizing theory, loosely categorizable as "objectivism", and the inherent situated-ness of personal experience, i.e. "perspectivism". [ HISTORICAL SKETCH OF WITH EXAMPLES]

Prior to the specialization of labour thinking and writing about these concerns was not strictly confined to a specific class or group of scholarship, be it natural science, philosophy, or the literary arts. Fortunately many honest and well executed supererogatory attempts to provide accommodating space for this underlying tension in the philosophy and practice of science persist. [ CONTEMPORARY SUMMARY OF EXAMPLES]

We write in this spirit and ask to be evaluated with respect to whether our work extends this narrative space insofar as (without detriment to any such philosophical or empirical stance in either the abstract or practice) diverse perspectives remain both expressible even if they cannot assert mutually intelligible.

A simple historical reading of these concerns:

"Some say the soul is mixed in with the whole universe, and perhaps this is why Thales

supposed that all things are full of gods.” (Aristotle, On the Soul 1.5 411a7-8)

“Since we have always recognized God’s wisdom in the detail of the mechanical structure of some particular bodies, it must also be displayed in the general economy of the world and in the constitution of the laws of nature. This is true to such an extent that once can observe the counsels of this wisdom in the laws of motion in general. For if there were nothing in bodies but extended mass and nothing in motion but change of place and if everything should and could be deduced solely from these definitions by geometrical necessity, it would follow, as I have shown elsewhere, that upon contact, the smallest body would impart its own speed to the largest body without losing any of this speed; and we would have to accept a number of such rules which are completely contrary to the formation of a system. But the decree of divine wisdom always to conserve the same total force and the same total direction has provided for this.” (Leibniz, Discourse on Metaphysics S21)

“The rationalists believed that we human beings can acquire important knowledge of reality by the use of our minds alone, by thinking, by pure reason. The empiricists denied this they insisted that experience was always a necessary ingredient and that all our knowledge of what actually exists must somehow or other be derived from experience. Again, the traditional view has been that these two opposing schools came together at the end of the eighteenth century and combined in the work of Immanuel Kant” (Bryan Magee, 1987)

I am trying to build useful tools, ideally composing a sufficient framework, for understanding dynamics such that the widest range of philosophical readings remain commensurable under their use. The dogma taken is that, under proper readings of modern phenomenology as providing the necessary synthesis for overly reductive epistemological challenges induced by "continental rationalism" vs "(british) empiricism", adequate reformulations of modern measure theoretic probability may be helpful for a number of sticky problems.

## TECHNICAL READING OF STUDY:

Current foundational research in probability is increasingly focused on moving beyond the classical Kolmogorov framework, which assumes a fixed sample space and builds models from coordinate-based descriptions. Much work today explores how probability should be defined when observations are contextual, limited, or structured by experimental constraints rather than by an underlying state space. This includes generalized probability theories, causal and contextual models, and operational approaches in quantum and statistical foundations. A key shift in these areas is toward treating observations, experiments, and information as primary objects, rather than assuming a predefined ontology of states or trajectories. A "philosophically secular" approach fits within this broader trend but with an attempt towards distinctive emphasis.

Instead of starting from coordinates or time-indexed variables, the idea is to treat queries as the primitive structure and construct probability measures by ensuring consistency across observations. This reframes classical stochastic processes as a special case of a more general observational completion principle. The optimistic upshot is describing how dynamical and Markov structures can emerge from empirical regularities. If successful a number of avenues might be possible including the investigation of formal and natural systems on their most general ground.

Of personal interest is: a more robust understanding of diffeomorphic embedding theorems (i.e. Takens) as a type of sufficient information geometric statistic; how to well-pose fundamental structural questions pertaining to the various epistemological desiderata one may adopt towards modern machine learning approaches (e.g. sufficiency, consistency, parsimony, etc.), and; non-equilibrium statistical mechanics.